

Change Data Management Service Project at kSynerX

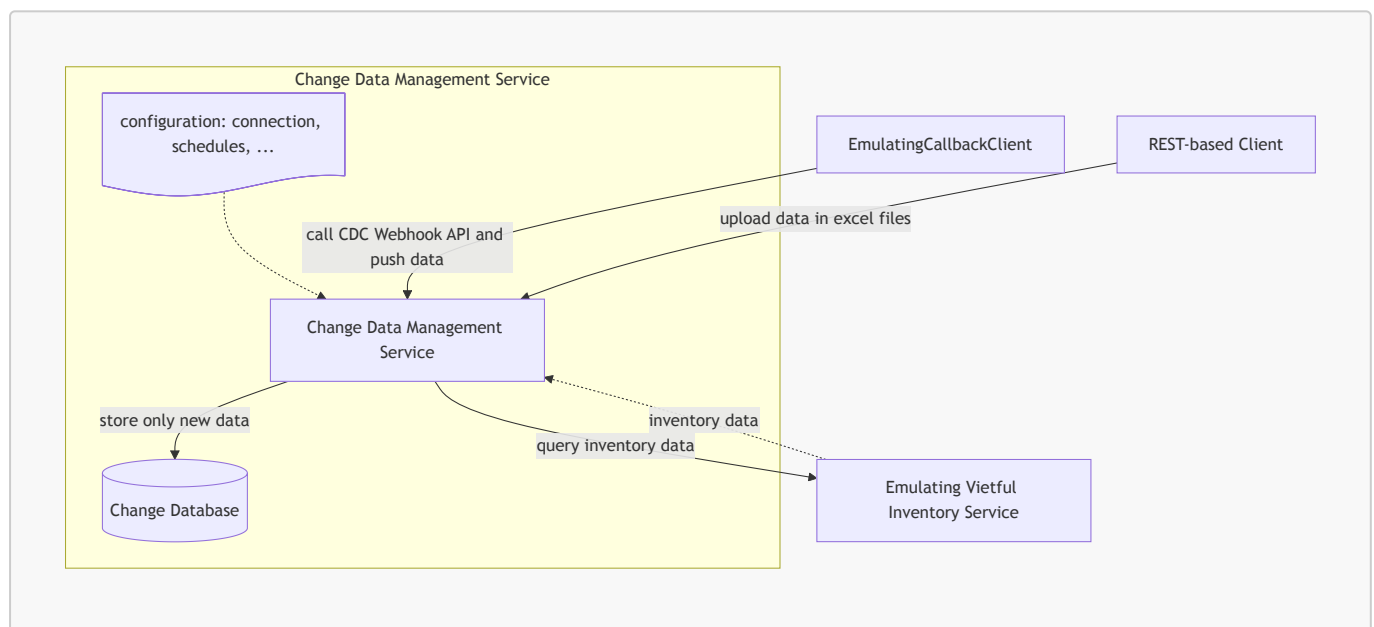
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The goal of this assignment is to develop a prototype of a simple Change Data Management Service (CDMS) for storing only changes of data, including:

- emulating a real inventory service (with limited features).
- developing a change data service with different update mechanisms.
- implementing exactly once change with possible failure handles.
- performing tests under spike loads.

Basic System Specification



- Emulating Vietful Inventory Service: based on the API described in: <https://ext.stg.vnfa.com/doc/index.html>.
- Database technology: PostgreSQL.
- Change Data Management Service (CDMS): a (composite) service supports three data processing mechanisms: querying data via pre-defined scheduled intervals, receiving data via webhook callbacks, and receiving data via excel files uploaded via REST API.
- Emulating data using faker (<https://faker.readthedocs.io/en/master/>).
- Data for management: the focus is for inventory product data (<https://ext.stg.vnfa.com/doc/index.html#tag/Products>). The goal is to capture new data about products (e.g., new products, changes of existing products).

Functionality requirements

- CDMS should be run with a single machine.
- Exactly once update: CDMS only stores new data into the database. No old data, no duplicated data are allowed.

- CDMS should work correctly under failures of CDMS, Vietful, databases, and machine hosting CDC.
- Deployment with a containerized environment.
- Test with spike and concurrent data processing mechanisms.

Assignment Outcomes

- Present your working prototype, with git code and documents in markdown.
- Demonstrate and explain your working prototype
- Present lessons learned.
- Clearly explain which features have been considered and implemented and which ones have not been finished.
- Clearly explain which code is written by the candidate and by using AI tools.